

BHARATH KUMAR POLABOINA

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Professional Summary

DevOps Engineer with hands-on experience in designing, automating, and maintaining cloud infrastructure on AWS. Experienced in developing Python-based automation, Infrastructure as Code using CloudFormation, Linux administration, Identity and Access Management (IAM), and serverless architectures. Proven ability to improve operational efficiency through automation, reducing manual effort from 90 minutes to 2 minutes. Passionate about building scalable infrastructure, improving system reliability, troubleshooting production issues, and automating operational workflows.

Technical Skills

Cloud Platforms	AWS (EC2, IAM, Lambda, S3, SQS, SES, CloudFormation, CloudWatch, RDS, ACM), Azure
Programming & Scripting	Python, Shell Scripting, SQL
DevOps & IaC	Docker, Kubernetes, Jenkins, Git, Terraform, Ansible, CloudFormation
Observability	Amazon CloudWatch, CloudWatch Alarms, Log Analysis, Metrics, Production Monitoring
Operating Systems	Linux, Windows
Methodologies	Serverless Computing, Event-Driven Architecture, CI/CD, REST APIs, IaC

Professional Experience

Cognizant	September 2024 – Present
DevOps Engineer	

- Designed and automated AWS infrastructure using CloudFormation templates, minimizing manual infrastructure provisioning and standardizing environment setups.
- Developed robust Python automation scripts for infrastructure management, operational workflows, and end-to-end deployment pipelines.
- Built scalable serverless applications utilizing Amazon Lambda, Amazon S3, Amazon SQS, and Amazon SES for high-throughput enterprise tasks.
- Designed an event-driven bulk email processing system capable of asynchronously processing thousands of email requests with fail-safe error handling.
- Configured granular IAM roles, policies, and resource-based permissions adhering strictly to least-privilege security principles.
- Implemented comprehensive CloudWatch monitoring, custom metrics, proactive alarms, and automated health-check processes for production workloads.
- Performed advanced Linux server administration, system troubleshooting, active process monitoring, and systematic log analysis.
- Investigated critical production incidents using CloudWatch Logs and AWS monitoring tools to isolate root causes and mitigate downtime.
- Collaborated closely with cross-functional application teams to troubleshoot complex infrastructure, deployment, and networking bottlenecks.
- Automated repetitive operational tasks across development and UAT environments, significantly reducing manual intervention and boosting team productivity.

Key Achievements at Cognizant:

- 97%+ Efficiency Gain:** Automated daily infrastructure health checks, drastically reducing manual verification effort from 90 minutes to under 2 minutes.
- Architected Scalable Email System:** Successfully designed and deployed an AWS serverless bulk email processing architecture (S3, Lambda, SQS, SES).
- Standardized Provisioning:** Created modular and reusable CloudFormation templates to accelerate environment provisioning across teams.

- Worked with Docker and Kubernetes for containerizing applications and orchestrating multi-container deployments.
- Assisted senior engineers in infrastructure automation, configuration management, and server provisioning using Ansible.
- Supported the continuous integration and delivery (CI/CD) pipeline implementation using Jenkins and Git version control.
- Gained hands-on practical experience with Azure Cloud deployments, resource group management, and infrastructure maintenance.

Projects

AWS Serverless Bulk Email Automation

Technologies: AWS Lambda, Amazon S3, Amazon SQS, Amazon SES, Python, CloudFormation

- Designed and implemented an event-driven serverless architecture that ingests CSV files uploaded to Amazon S3 and automatically triggers bulk email dispatch via Amazon SES.
- Developed core Python Lambda functions to handle data parsing, error validation, and asynchronous SQS queue message creation.
- Configured SQS batch processing and Event Source Mapping to optimize throughput, ensure reliable message delivery, and implement automatic retries with Dead Letter Queues (DLQ).
- Implemented fine-grained IAM roles and resource security policies to restrict cross-service access.

CloudWatch Infrastructure Health Monitoring Automation

Technologies: Python, Amazon CloudWatch, AWS CloudFormation, Linux Shell

- Architected and automated an infrastructure health monitoring system replacing slow, manual server inspection processes.
- Developed Python scripts to capture server process health, system metrics, and log anomalies, pushing custom metrics directly to Amazon CloudWatch.
- Configured CloudWatch Alarms to trigger automated notifications for early incident detection, cutting monitoring overhead from 90 to 2 minutes.

Education

Bachelor of Technology in Information Technology

Chaitanya Bharathi Institute of Technology

Graduated

CGPA: 8.59 / 10.0

Certifications & Key Recognition

- **AWS Certified Cloud Practitioner** — Amazon Web Services
- **Microsoft Certified: Azure Fundamentals (AZ-900)** — Microsoft
- **Cognizant Management Award:** Recognized and awarded by management for delivering an infrastructure automation solution that reduced operational effort by over 97%.
- **Top 5 Competitive Programmer:** Ranked among the Top 5 institute-level problem solvers on GeeksforGeeks; recipient of GeeksforGeeks official recognition and rewards.